FINAL-YEAR RESEARCH / THE IDEA AT A GLANCE

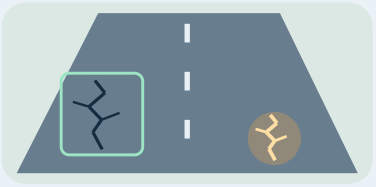
Road damage across countries

Find more damage. Flag where the detector may have missed it.

YOLO = the AI detector that draws boxes around damage.

THE PROBLEM

Roads, cameras and lighting change between countries. An AI detector trained elsewhere may miss small or thin damage.



THE FAIR TEST

TRAIN ON 3 COUNTRIES
Learn only from source data

→

FREEZE THE MODELS
No tuning on the test country

→

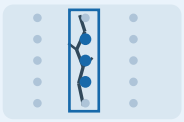
TEST ON THE 4TH
Measure unseen-country results

Rotate the test country: Japan · India · US · Czech Republic

THREE IDEAS THE TEAM WILL TEST

A MEMBER A

Learn from thin boxes



Which examples teach the model?

Changes how YOLO chooses and weights training examples for thin damage.

DEEP LEARNING: retrains YOLO

Compare: An existing assignment rule (plain NWD), tuned equally.

MEASURE: detection accuracy, especially on small/thin boxes.

B MEMBER B

Compare two image scales

Full view · Small view

Trains YOLO to make consistent predictions from a full-size and smaller view.

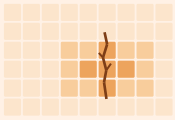
DEEP LEARNING: trains YOLO

Compare: Consistency for small/thin damage vs all damage; random-subset check.

MEASURE: how much small/thin damage is found, and false alarms.

C MEMBER C

Predict likely miss locations



Likely miss - not a new box

Uses YOLO's visual features to train a small neural network. YOLO stays unchanged.

DEEP LEARNING: trains a neural head

Compare: Visual features + weak guesses vs a model using weak guesses alone.

MEASURE: how well warnings identify locations with missed damage.

ONE BASELINE. THREE SEPARATELY TESTED CONTRIBUTIONS.

A and B aim to improve detection. C aims to reveal likely misses.

PUBLIC DATA
RDD2022; existing labels

REPEAT RUNS
3 seeds for main tests

FAIR CONTROLS
Baselines + ablations

These are proposed methods: improvements must be demonstrated, not assumed.

Based on proposal v8. Dataset: Arya et al., RDD2022 (2024), DOI 10.1002/gdj3.260.